

Lab 1: Requirements Engineering & UML Use-Case Modelling

Deliverable 3 — Use-Case Flow Specification

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Course	Software Engineering (UE24CS341A) Semester 5 Section F Dr Ashok Kumar Patil	Problem Statement	#65 — Sustainability & Green Tech: Urban Rooftop Gardening Co-op App		

1. Use-Case Identification

Use Case ID	UC-05
Use Case Name	Trade Surplus Produce
Related Requirements	FR-003, FR-004, NFR-001
Primary Actor	Co-op Gardener (buying member)
Secondary Actors	Co-op Gardener (listing member), Payment Gateway
Stakeholders & Interests	Buying gardener - wants surplus produce at a fair price with a guaranteed pickup; Listing gardener - wants surplus sold before spoilage with assured settlement; Community Leader - wants an auditable, waste-reducing trade record.
Trigger	The buying gardener selects a surplus listing on the community trade board.
Preconditions	1. Both gardeners are registered co-op members and the buyer is authenticated. 2. The listing gardener's rooftop plot status is "Active" (approved under FR-002). 3. The surplus listing is published, is inside its availability window and has quantity > 0 kg. 4. The Payment Gateway is reachable.
Postconditions (success guarantee)	1. The listed quantity is decremented by the traded amount. 2. A trade record with trade ID, authorisation code, quantity and pickup window is persisted. 3. Receipts are issued to both gardeners and the trade board is refreshed for all members within 2 s (NFR-001).
Postconditions (minimal / failure guarantee)	No quantity is decremented, no funds are captured, any soft reservation is released back to the listing, and the failed attempt is logged for audit.

2. Main Success Scenario

Step	Actor / System action
1	The buying gardener opens the Produce Trade Board; the system displays all active surplus listings with crop, available quantity, exchange mode and pickup location.
2	The buying gardener selects a listing and enters the quantity, in kilograms, to reserve.
3	The system verifies the buyer's co-op membership and session («include» UC-07 Verify Member Credentials) and checks that the requested quantity does not exceed the available quantity.
4	The system places a soft reservation on the requested quantity, valid for 10 minutes, and displays the trade summary with the payable amount and the pickup window.
5	The buying gardener selects the exchange mode - sale payment or barter confirmation - and confirms the trade.
6	The system invokes Process Trade Payment («include» UC-10), forwarding the trade ID and amount to the Payment Gateway. Where the buyer is an eligible member, Apply Co-op Member Discount («extend» UC-11) reduces the payable amount before authorisation.
7	The Payment Gateway authorises the payment and returns an authorisation code.
8	The system decrements the listing quantity, converts the soft reservation to "Confirmed" and persists the trade record.
9	The system issues a receipt carrying the trade ID and pickup window to both gardeners, notifies the listing gardener, and refreshes the trade board for all community members within 2 s.
10	The use case ends successfully.

3. Alternate Flows

Alternate Flow 7a - Payment Declined
7a1. The Payment Gateway returns a "declined" response with a reason code.
7a2. The system displays a "Payment failed" message, retains the soft reservation for its remaining validity, and prompts the buying gardener to choose another payment method.
7a3. The buying gardener selects an alternative payment method and the use case resumes at step 6.
7a4. If the payment is declined again (after 2 attempts in total), the system cancels the reservation, restores the quantity to the listing, notifies both gardeners that the trade was not completed, and logs the failed attempt. The use case ends in failure.
Alternate Flow 4a - Quantity Already Claimed
4a1. At step 4 the system finds that the requested quantity now exceeds the available quantity because another member reserved it concurrently.
4a2. The system displays the updated available quantity and offers the buying gardener the choice of reserving the reduced quantity or joining the waiting list for that crop.
4a3. If a reduced quantity is accepted the use case resumes at step 2; otherwise no reservation is created and the use case ends without a trade.

Traceability: UC-05 realises FR-003 and FR-004; step 3 exercises the «include» relationship with UC-07, step 6 the «include» relationship with UC-10 and the «extend» relationship with UC-11, and step 9 is bounded by the 2-second trade-board refresh of NFR-001.